



# Crush Syndrome Trauma

## History

- Entrapped and crushed under heavy load > 30 minutes
- Extremity / body crushed
- Building collapse, trench collapse, industrial accident, pinned under heavy equipment

## Signs and Symptoms

- Hypotension
- Hypothermia
- Abnormal ECG findings
- Pain
- Anxiety

## Differential

- Entrapment without crush syndrome
- Vascular injury with perfusion deficit
- Compartment syndrome
- Altered mental status

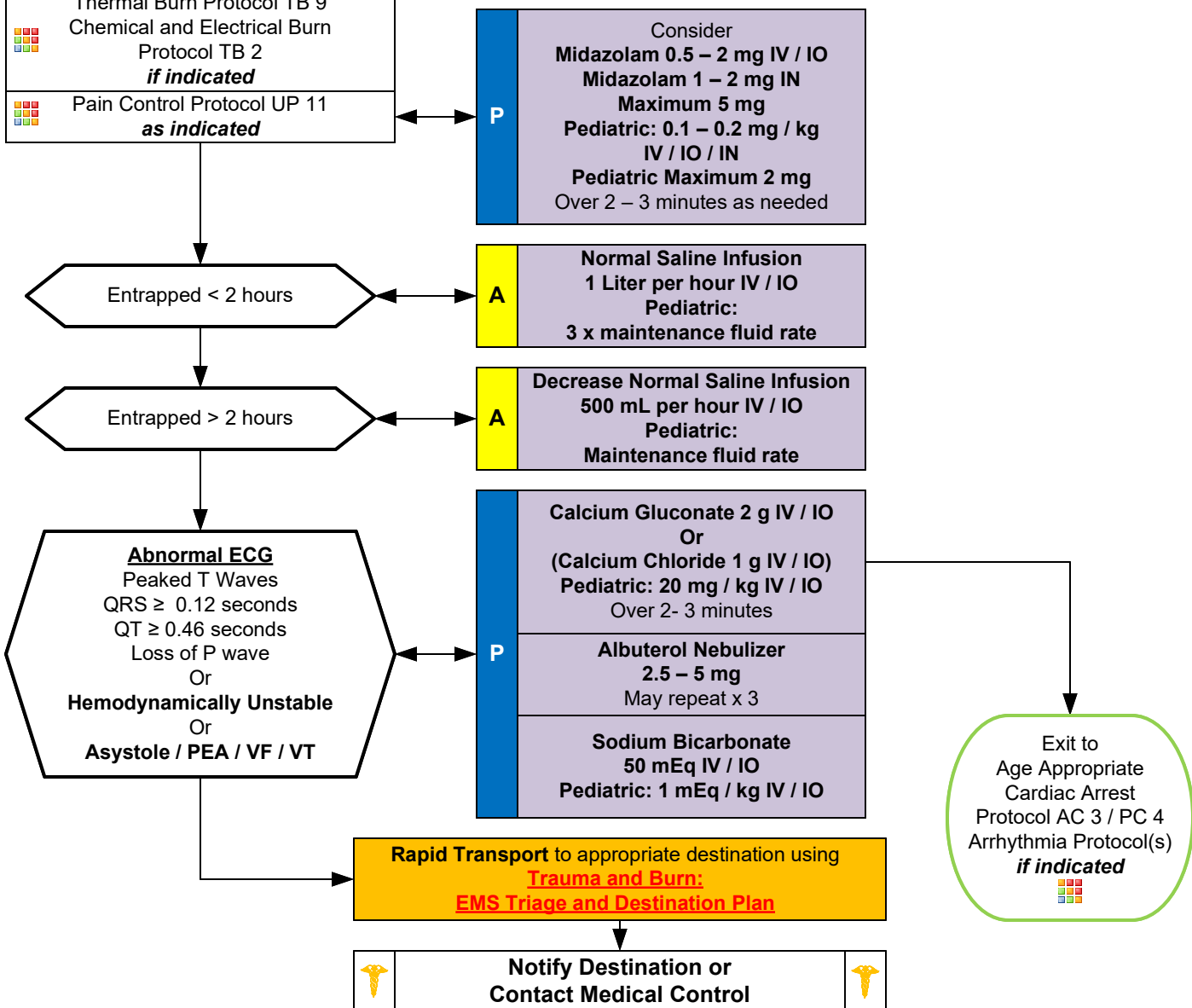
|          |                                                                                                    |
|----------|----------------------------------------------------------------------------------------------------|
|          | Age Appropriate<br>Airway<br>Protocol(s) AR 1 - 7<br><b>as indicated</b>                           |
| <b>B</b> | 12 Lead ECG Procedure                                                                              |
|          | IV or IO Access Protocol UP 6                                                                      |
| <b>P</b> | Cardiac Monitor                                                                                    |
|          | Multiple Trauma Protocol TB 6<br><b>if indicated</b>                                               |
|          | Thermal Burn Protocol TB 9<br>Chemical and Electrical Burn<br>Protocol TB 2<br><b>if indicated</b> |
|          | Pain Control Protocol UP 11<br><b>as indicated</b>                                                 |

### Age Specific Blood Pressure indicating possible shock

Age 0 – 28 days: SBP < 60  
Ages ≥ 1 month: SBP < 70  
Age 1 – 9: SBP < 70 + (2x Age)

Ages 10 – 64: SBP < 90  
Ages ≥ 65: SBP < 100

All ages:  
Shock Index HR > SBP





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## Pearls

- **Recommended exam: Mental Status, Musculoskeletal, Neuro**
  - **Scene safety is of paramount importance as typical scenes may pose hazards to rescuers. Call for appropriate resources.**
  - **Crush Injury is a localized crush injury with systemic signs and symptoms causing muscle breakdown and release of potentially toxic muscle cell components and electrolytes into the circulation.**
  - **Crush syndrome typically manifests after 1 – 4 hours of crush injury.**
  - **Fluid resuscitation strategy:**
    - If possible, administer IV / IO fluids prior to release of crushed body part, especially with crush > 1 hour.
    - If access to patient and initiation of IV / IO fluids occurs after 2 hours, give 2 liters of IV fluids in adults and 20 mL/kg of IV fluids in pediatrics, and then begin > 2 hour dosing regimen.
  - **If not able to perform IV / IO fluid resuscitation immediately, place tourniquet on crushed limb until IV / IO fluids can be initiated (even if tourniquet is not being used for hemorrhage control).**
  - **Pediatric IV Fluid maintenance rate:**
    - 4 mL for the first 10 kg of weight +
    - 2 mL for the second 10 kg of weight +
    - 1 mL for every additional kg in weight after 20 kg
- Example: 28 kg pediatric**

|                      |                       |
|----------------------|-----------------------|
| First 10 kg:         | 4 mL/kg/hr = 40 mL/hr |
| Second 10 kg:        | 2 mL/kg/hr = 20 mL/hr |
| Final 8 Kg:          | 1 mL/kg/hr = 8 mL/hr  |
| Total: 68 mL/hr rate |                       |
- **Consider all possible causes of shock and treat per appropriate protocol.**
  - **Majority of decompensation in pediatrics is airway or respiratory related.**
  - **Decreasing heart rate and hypotension occur late in children and are signs of impending cardiac arrest.**
  - **Shock may be present with a normal blood pressure initially or even elevated.**
  - **Shock often is present with normal vital signs and may develop insidiously. Tachycardia may be the only sign.**
  - **Patients may become hypothermic even in warm environments. Maintain warmth.**
  - **Hyperkalemia from crush syndrome can produce ECG changes described in protocol, but may also be a bizarre, wide complex rhythm. Wide complex rhythms should also be treated using the VF/ Pulseless VT Protocol if indicated (AC 9 VF Pulseless VT Protocol and/ or PC 7 Pediatric VF Pulseless VT Protocol).**